



Quantitative and classification analysis of red wine by infrared spectra and gas chromatography–mass spectrometry data coupling with a new variable selection method

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ABSTRACT

A new chemometric variable selection model named as consensus G-based variable selection (G-based VS) was constructed. The model combined consensus strategy with uncorrelated linear discriminant analysis to select characteristic variables based on "load capacity" ($\bar{G}$ value) and occurrence frequency of each variable in multiple sub-models (member models). It achieved accurate classification results of wine from different geographical origins with multi-type datasets of infrared spectra, chromatography and mass profile. Then IR spectra combined with least squares support vector regression (LSSVR) was utilized for the prediction of multiple quality parameters of red wine with the selected feature variables, and the results showed good performance in terms of prediction accuracy and robustness compared with other published variable selection methods. Therefore, the proposed method shows potential in variable selection for the quantitative and classification analysis of red wine.

1. Introduction

Nowadays, red wine is becoming increasingly popular in daily life. Many studies have focused on the wine quality parameters evaluation (lactic acid, total polyphenol index, pH and volatile acid etc.) based on experimental methods including titration, colorimetry, distillation, chromatography, gas chromatography–mass spectrometry, etc. Besides, the authenticity identification of the geographical origin of grapes and wines is also important for ensuring wine quality. Isotope ratio mass spectrometry (IRMS)-based, ICP-MS-based, UPLC–QTOF-MS-based metabolomic, and stable isotope techniques are available for the geographical origin identification and tracking of red wines (Gao et al., 2022; Fan et al., 2018; Wang et al., 2023; Pan et al., 2022), which obtain wine fingerprint information and construct wine fingerprint spectrum systems. This work introduced chemometrics to experimental techniques of infrared spectra (IRs) and chromatography; with the help of chemometric modeling, samples feature extraction and simultaneous multi-component determination could be quickly achieved. Furthermore, the subjectivity of artificial comparison could be avoided.

As the basic theory and methodology of chemical measurement,

chemometrics is playing an increasingly important role in the analysis of sophisticated samples (Genisheva et al., 2018; Abdul et al., 2020; Ma et al., 2020; Beganović et al., 2020). It helps establish the relationships between the measurement data and the content or categories of multiple components in the sample (Liu et al., 2019; Ming et al., 2021; Yun et al., 2019; Bell et al., 2017; Long et al., 2023). In-depth extraction of the sample characterized information and replacement of "chemical separation" with "mathematical separation" allow simultaneous determination of multiple components or parameters without affecting the predicted results in the presence of unknown interferences (Liang et al., 2016).

In chemometric multivariate modeling, the increasing amount of high-dimensional and massive data presents new challenges for the extraction of characterized information in sophisticated samples. Not all variables contribute to the model, and therefore the extraction of information variables has become one of the general and critical issues requiring more research. It has been proved that identifying and removing the uninformative variables can simplify the model and improve the prediction performance of the model (Li et al., 2021; Asante et al., 2022; Wang et al., 2022). Thus, many variable selection methods

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Table 1

The geographical origin of the red wines.

Origin	Argentina	Chile	Australia	South Africa	Total
Wine samples	6	15	12	11	44

emerge, such as uninformative variable elimination (UVE) (Centner et al., 1996), Monte-Carlo uninformative variable elimination (MC-UVE) (Cai et al., 2008), variable importance in projection (VIP) (Galindo-Prieto et al., 2017), interval PLS (IPLS) (Norgaard et al., 2000), competitive adaptive reweighted sampling (CARS) and so on (Hu et al., 2018; Zhang et al., 2022; Forina et al., 2015; Figueiredo et al., 2018; Zhang et al., 2019a). It is found that many common methods are based

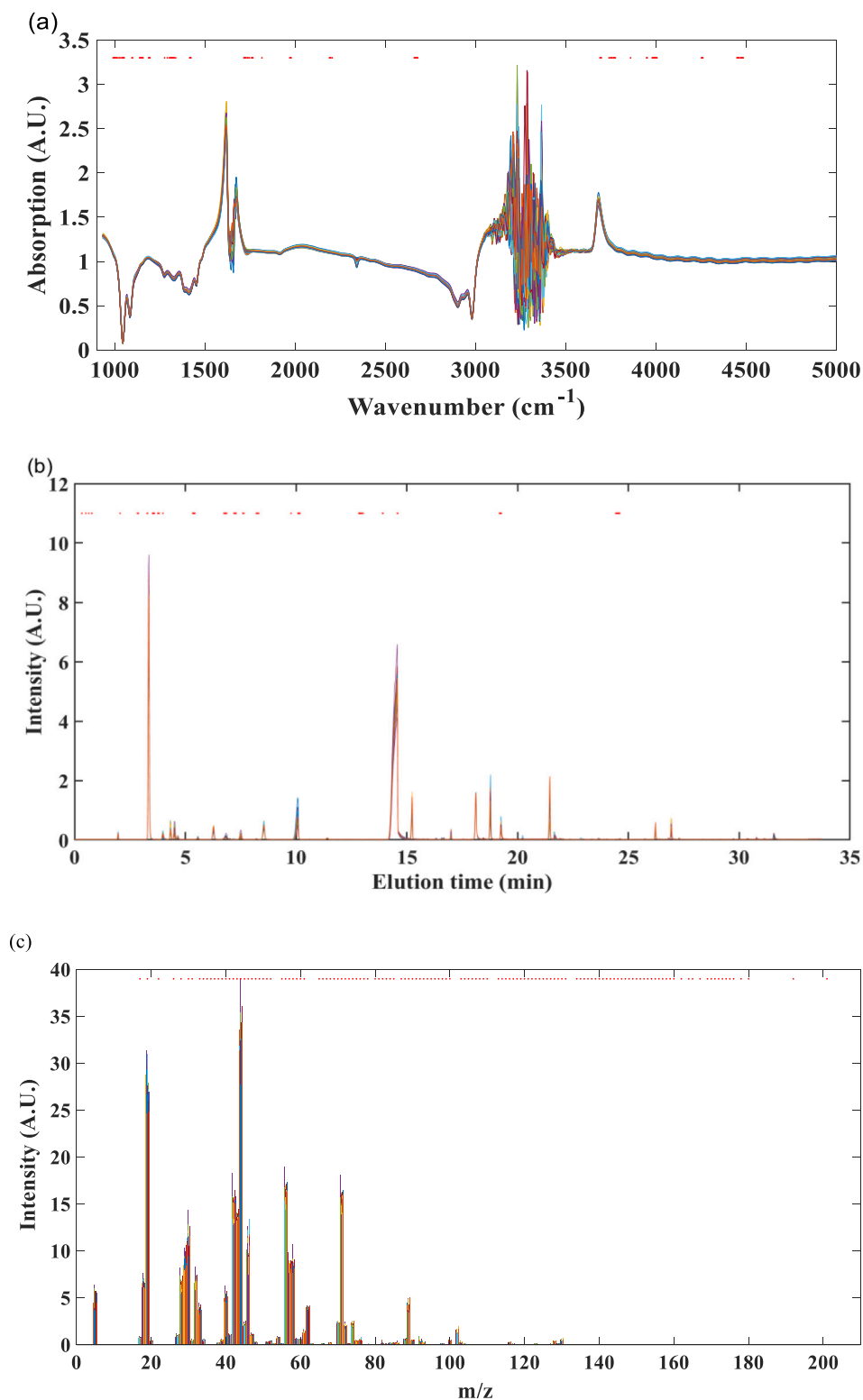
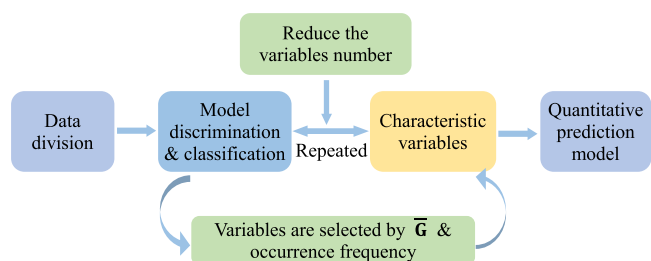


Fig. 1. The samples original graph (bottom) and scatter diagram of the selected variables (upper) (a: IR spectra; b: chromatography; c: Mass spectrometry).

Table 2

Minimum, maximum, mean and standard deviation (σ) values of the red wine parameters.

Parameters	Minimum	Maximum	Average	Standard deviation (σ)
Ethanol (vol%)	12.79	15.09	13.92	0.47
Total acid (g/L)	2.99	4.78	3.73	0.48
Volatile acid (g/L)	0.21	1.03	0.40	0.14
pH	3.48	3.92	3.69	0.09
Lactic acid (g/L)	0.78	2.95	1.11	0.37
Density (g/mL)	0.99	1.00	0.99	0.00
Total polyphenol index	44.68	83.51	61.89	7.07
Glycerol (g/L)	8.12	12.52	10.20	0.90
Methanol (vol%)	0.13	0.30	0.20	0.04

**Fig. 2.** The flow chart of consensus $\bar{G}$ -based variable selection model.

on the partial least squares (PLS) regression coefficients to select informative variables. Nevertheless, the values of the regression coefficients are possibly influenced by data collinearity, improper pretreatment method, or fluctuating from the background, which can lead to prediction error.

In this study, a new variable selection model called consensus $\bar{G}$ -based variable selection (called “ $\bar{G}$ -based VS” for short) based on consensus strategy (Li and Jing, 2014) combined with uncorrelated linear discriminant analysis (ULDA) was constructed to deal with three red wine datasets of infrared spectra, mass spectrometry and chromatography. The model allows fast and accurate identification of the wine from different geographical origins, and simultaneously selects characteristic variables from a large number of variables. Furthermore, with the characteristic variables, IRs combined with least squares support vector regression (LSSVR) was successfully utilized to predict multiple principal quality parameters of the red wine. It is noted that the purpose of comparing alternative methods in this work is to demonstrate that $\bar{G}$ -based VS is a valid method for variable selection and to provide new insights into variable selection, rather than to evaluate which method is better.

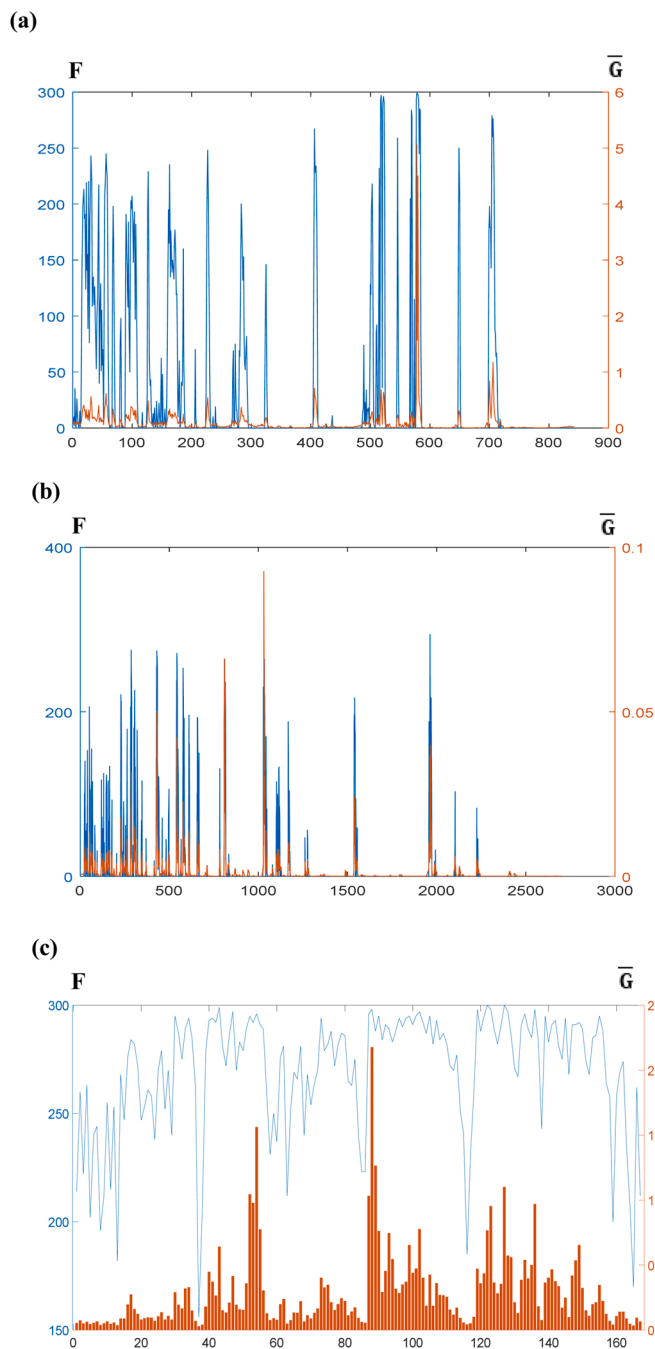
2. Theory and algorithm

2.1. Wavelet transform preprocessing method

The wavelet transform (WT) is a mathematical transform that performs hierarchically decomposing functions, and is based on the Fourier transform (Ramin et al., 2022; Mishra et al., 2020; Shao and Cai, 1998; Tian et al., 2023). After research by Daubechies (1988) and Mallat (1989), and other researchers, real breakthroughs have been made in the application of the WT in the analysis and processing of various signals. Discrete wavelet transform (DWT) is a special case of WT, which provides a compact representation of a signal in time and frequency, and can efficiently eliminate noise and other redundant information.

2.2. Uncorrelated linear discriminant analysis (ULDA)

Uncorrelated linear discriminant analysis (ULDA) has been an

**Fig. 3.** The $\bar{G}$ (—) and occurrence frequency (F, —) of each variable (a: IR spectra; b: chromatography; c: mass spectrometry).

algorithm of feature extraction and dimension reduction based on discriminant analysis, which was first proposed by Jin et al. (2001) to extract optimal discriminant features for face images. Unlike Fisher discrimination analysis, ULDA attempts to find a transformation matrix $\bar{G}$ that satisfies “s-orthogonality” by using singular value decomposition as a tool. The original high-dimensional variables can be converted into the low-dimensional independent variables (uncorrelated discriminant vector, UDV) through “load capacity” $\bar{G}$. It takes into account the uncorrelatedness between column vectors, where each column in the low-dimensional matrix is a linear combination of all the columns in the original high-dimensional matrix. As a result, UDV with less information redundancy and higher discriminant ability can be calculated by the formula (1). Further details can be found in the literature (Liang et al., 2016; Li and Zeng, 2016).

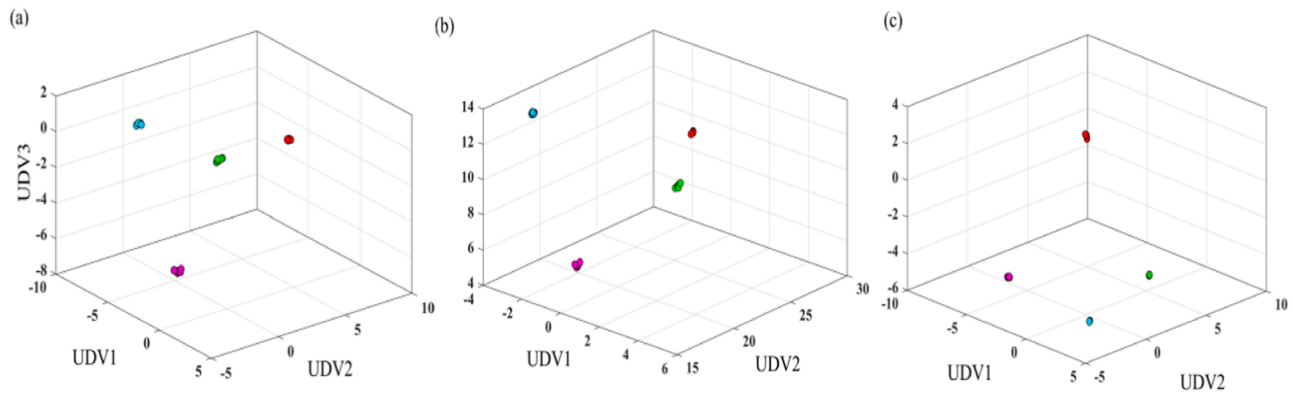


Fig. 4. The identification results of IR spectra data (●: Argentina; ●: South Africa; ●: Chile; ●: Australia) (n : The number of retained variables. a: $n = 489$, b: $n = 192$, c: $n = 96$).

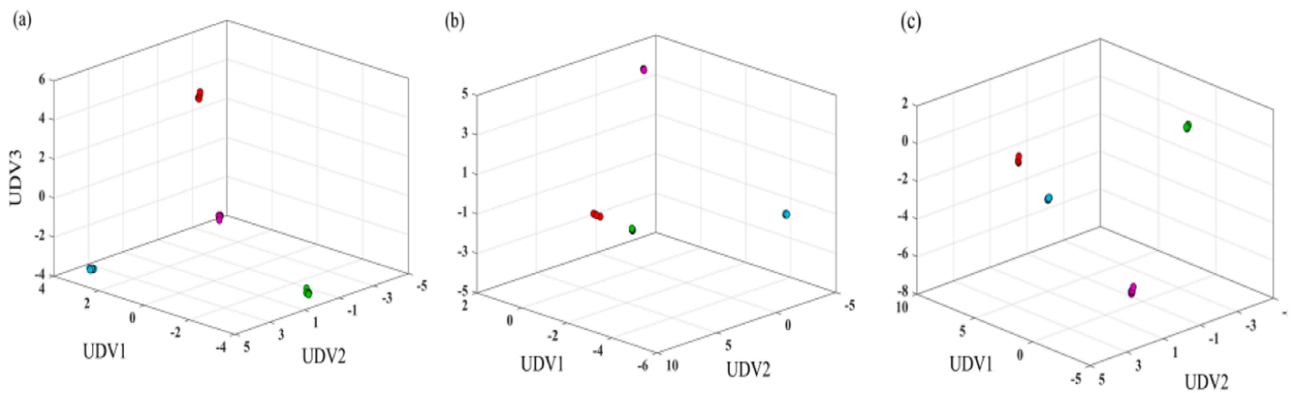


Fig. 5. The identification results of chromatography data (●: Argentina; ●: South Africa; ●: Chile; ●: Australia) (n : The number of retained variables. a: $n = 989$, b: $n = 481$, c: $n = 92$).

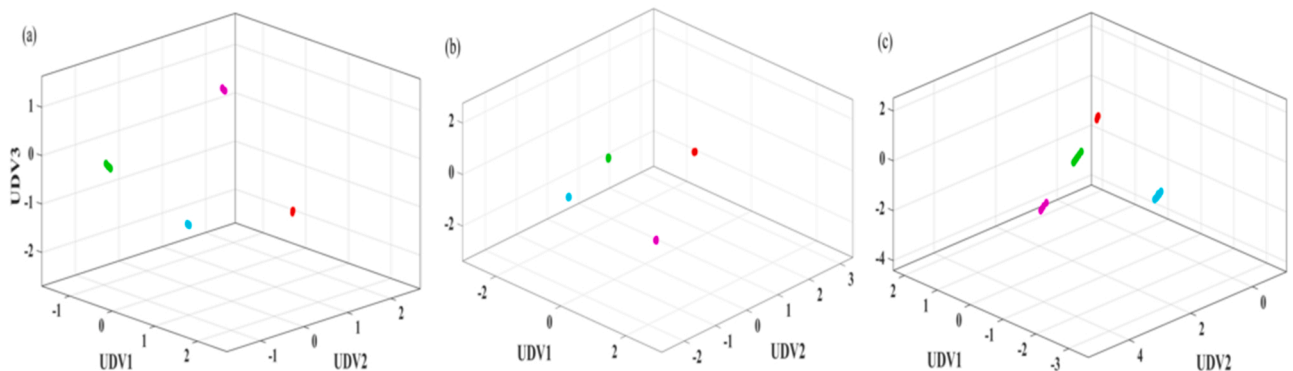


Fig. 6. The identification results of mass spectrometry data (●: Argentina; ●: South Africa; ●: Chile; ●: Australia) (n : The number of retained variables. a: $n = 167$, b: $n = 148$, c: $n = 138$).

$$\text{UDV} = \text{XG} \quad (1)$$

where X is the original data matrix, and G is the transformation matrix.

2.3. Least squares support vector regression (LSSVR)

The principle of the support vector regression (SVR) lies in the non-linear transformation of the initial data, mapping it from the original space to another high-dimensional feature space by kernel functions, and constructing the best linear regression function in this space. Least squares support vector regression (LSSVR) is a transformation of SVR,

Table 3

The classification results of different methods for the red wine.

Method	full PLS- DA	BSS	EBSS	GA- PLSDA	s- PLSDA	G-based VS
Classification accuracy (%)	59.3	60.0	71.1	60.4	66.8	100

which constructs the Lagrangian function as follows to solve the linear system,

$$\begin{bmatrix} 0 & \mathbf{I}_n^T \\ \mathbf{I}_n & \mathbf{K} + \gamma^{-1}\mathbf{I} \end{bmatrix} \begin{bmatrix} b_0 \\ \mathbf{b} \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbf{y} \end{bmatrix} \quad (2)$$

where $\mathbf{K}$ is a kernel matrix, $\mathbf{I}_n$ is a $[n \times 1]$ vector, $\mathbf{T}$ is a transpose of a matrix or vector, γ is a weight vector, b_0 is the model offset, and $\mathbf{b}$ is a regression vector. LSSVR improves the calculation efficiency of SVR, and shows excellent suitability for nonlinear datasets of small samples. More details of the LSSVR algorithm are published (Li et al., 2007; Pakniyat et al., 2021). In this study, Gaussian Radial Basis Function (RBF) kernel function was used with a kernel width parameter σ of 2, and the optimized weight vector γ is set to 10^6 .

3. Experimental and calculations

3.1. Experimental and data

The red wine dataset is provided by http://www.models.life.ku.dk/Wine_GCMS_FTIR. Red wines (44 samples) were produced from the same grape (100% Cabernet Sauvignon) and harvested in four different geographical areas. Details on the geographical origins and number of wine samples analyzed are given in Table 1.

The wine sample data was obtained by using headspace GC-MS and FT-IR analytical instruments. FT-IR spectra were scanned from 929 to 5011 cm^{-1} at 4 cm^{-1} intervals. Water band around 1545–1710 cm^{-1} and 2968–3620 cm^{-1} were excluded for data analysis in this work (dimension: 44×842). For each sample, a mass spectrum scan measured at 2700 elution time-points was obtained providing a data cube of size $44 \times 2700 \times 200$. The original graphs of the red wine samples are shown in Fig. 1. Fig. 1(a) is the IR spectra with water band (dimension: 44×1056). Fig. 1(b) is the chromatogram showing the TIC (total ion count), in which the intensity at each scan is found by summing the contribution of all intensities of mass channels investigated (dimension: 44×2700). Fig. 1(c) demonstrates the mass profiles (summed elution time dimension). Due to the original data (dimension: 44×200) embracing the uninformative variables (intensity=0), these had been firstly removed, and the remaining variables (dimension: 44×167) were used.

Principal quality parameters of wine samples containing ethanol (vol %), total acid (g/L), pH, volatile acid (g/L), lactic acid (g/L), density (g/mL), total polyphenol index, glycerol (g/L), and methanol (vol%) were used to evaluate the model performance. Mean, minimum, maximum, and standard deviation values of the quality parameters are presented in Table 2.

3.2. Calculations

The process of consensus G-based variable selection (G-based VS) model is shown in Fig. 2.

- (1) Firstly, standardization and DWT were chosen to preprocess the original data. Through comparisons of different wavelet filters including "sym" (Symlets wavelet filter), "db" (Daubechies wavelet filter), "coif" (Coiflets wavelet filter) and decomposition

scale, "db10" filter and decomposition scale 5 were finally adopted.

- (2) Construct consensus G-based variable selection model: Consensus modeling was realized through k -fold cross-validation ($k = 2$) combined with the ULDA method. ULDA helps to find the best classification of subspace with characteristic variables. Along with gradually reducing the number of remaining variables, $\bar{\mathbf{G}}$ (the mean value of $\mathbf{G}$) and occurrence frequency of each variable in one hundred sub-models (member models) were sorted in descending order, and variables with both high Gvalue and high occurrence frequency were selected as characteristic variables. Then characteristic variables substituted the original variables to carry out the following classification and quantitative calculations.
- (3) In the classification analysis, according to the ULDA theory, $m-1$ uncorrelated discriminant vectors are obtained, where m represents the number of sample categories. In this work, m is four, so and three transformation matrices $\mathbf{G}_1$, $\mathbf{G}_2$, $\mathbf{G}_3$, and corresponding UDV_1 , UDV_2 , UDV_3 were obtained. Thus, $\mathbf{G}$ is defined as follows:

$$\mathbf{G} = \sqrt{((\mathbf{G}_1)^2 + (\mathbf{G}_2)^2 + (\mathbf{G}_3)^2)} \quad (3)$$

- (4) In the quantitative modeling, all samples were randomly divided into the training set, validation set, and prediction set in a ratio of 3:1:1. Root mean squared error of the validation set (RMSEV) is used as the model evaluation criterion. Root mean squared error of the prediction set (RMSEP), sample recovery, the determination coefficient (R), and p value are counted and compared.

$$\text{RMSE} = \left[\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \right]^{\frac{1}{2}} \quad (4)$$

$$\text{Recovery} = \hat{y}_i / y_i * 100\% \quad (5)$$

where $\hat{y}_i$ is the predicted value of the i th sample, y_i is the true value of the i th sample, and n is the number of the predicted samples.

4. Results and discussions

4.1. The results of variables selection

On the premise of ensuring the classification effect, the G-based VS model can select fewer informative variables to construct a more concise and effective model. The $\bar{\mathbf{G}}$ value and occurrence frequency ($\mathbf{F}$) in the sub-models of each variable are presented in Fig. 3. Clearly, for the three datasets, the fluctuation trends of $\bar{\mathbf{G}}$ and $\mathbf{F}$ are similar. In other words, most variables with higher $\bar{\mathbf{G}}$ values are just those with higher $\mathbf{F}$ value, which shows the rationality and robustness of the characteristic variable selection model. The corresponding variables with both higher $\bar{\mathbf{G}}$ values ("load capacity") and higher $\mathbf{F}$ values (occurrence frequency) were selected as important and characteristic variables for further classification and quantitative calculations.

4.2. The identification and classification results

Typical identification results from three datasets are shown in 3D Figs. 4–6, where uncorrelated discriminant vector values of the prediction set are given. The variables in the new low-dimensional matrix UDV converted from the original matrix are not correlative to each other, which is beneficial for feature extraction and minimizing information redundancy and interference.

As can be seen from 3D figures, at a different number of reserved variables, four groups cluster very obviously (near inter-class distance and far between-class distance), which indicates that the UDVs can

Table 4

The results of different variable selection methods for IR spectra data.

Parm.	Item	CARS-LSSVR	UVE-LSSVR	IPLS-LSSVR	MWPLS-LSSVR	FD-VS-LSSVR	G-based-LSSVR	Full-LSSVR
Ethanol	N_m	18(13.37) ^e	110(13.36) ^e	84	110	124	96	842
	RMSEP (%)	0.2293(0.037) ^e	0.2921(0.011) ^e	0.2980	0.3845	0.2175	0.2389	0.2471
	Recovery (%)	97.48–102.64	96.12–102.92	94.33–101.60	96.70–104.53	97.76–101.89	97.22–102.24	97.76–102.36
	R	0.5811	0.6488	0.8395	0.6237	0.8649	0.8363	0.8688
	p	≈ 0	≈ 0	0.0046	0.0727	0.0021	0.0050	0.0024
Total acid	N_m	37(23.61) ^e	480(11.68) ^e	85	91	109	96	842
	RMSEP (g/L)	0.1490(0.028) ^e	0.2299(0.015) ^e	0.3762	0.2928	0.2401	0.2558	0.2599
	Recovery (%)	95.68–108.57	93.60–110.79	79.95–102.07	91.42–119.60	98.71–115.86	89.15–111.41	95.75–111.26
	R	0.9703	0.9386	0.8591	0.8659	0.9363	0.9285	0.9111
	p	≈ 0	≈ 0	0.0030	0.0025	0.0002	≈ 0	≈ 0
Volatile acid	N_m	34(16.05) ^e	151(31.79) ^e	84	108	76	96	842
	RMSEP (g/L)	0.0367 (0.007) ^e	0.0431(0.007) ^e	0.0854	0.0720	0.0542	0.0449	0.0863
	Recovery (%)	87.04–118.62	77.21–118.70	66.80–136.49	78.51–118.81	62.68–121.73	77.91–111.10	58.47–133.31
	R	0.9577	0.9465	0.5118	0.6786	0.9770	0.9410	0.8002
	p	≈ 0	≈ 0	0.1948	0.0063	≈ 0	≈ 0	0.0096
pH	N_m	46(49.57) ^e	67(13.50) ^e	84	81	124	96	842
	RMSEP	0.0282(0.005) ^e	0.0467(0.017) ^e	0.0693	0.0432	0.0467	0.0446	0.0939
	Recovery (%)	98.66–100.93	97.74–101.72	99.10–103.87	98.02–101.46	97.62–100.36	97.72–101.26	97.80–100.96
	R	0.9629	0.8862	0.7940	0.9087	0.9385	0.9230	0.9454
	p	≈ 0	≈ 0	0.0106	0.0007	0.0002	≈ 0	≈ 0
Lactic acid	N_m	23(12.03) ^e	85(24.88) ^e	85	77	124	96	842
	RMSEP (g/L)	0.0722(0.009) ^e	0.1037(0.009) ^e	0.1203	0.0886	0.0800	0.0954	0.1611
	Recovery (%)	89.37–112.418	83.02–118.02	85.07–112.15	83.12–113.75	86.44–107.62	82.89–107.83	70.37–123.52
	R	0.9646	0.9240	0.8505	0.9454	0.9676	0.9654	0.8629
	p	≈ 0	≈ 0	0.0037	0.0001	≈ 0	≈ 0	0.0027
Density	N_m	26(11.20) ^e	414(24.64) ^e	85	29	59	96	842
	RMSEP (g/mL)	0.0036(0.001) ^e	0.0043(0.001) ^e	0.0067	0.0032	0.0043	0.0060	0.0048
	Recovery (%)	99.39–100.37	99.21–100.50	98.73–101.03	99.48–100.45	99.17–100.41	98.81–100.58	99.20–100.55
	R	0.7523	0.6919	0.6589	0.7897	0.7574	0.7180	0.7520
	p	≈ 0	≈ 0	0.0536	0.0114	0.0181	0.0294	0.0194
Total polyphenol index	N_m	20(13.58) ^e	571(20.07) ^e	85	83	218	96	842
	RMSEP	3.3051(0.359) ^e	3.8839(0.043) ^e	5.3127	5.2470	4.0888	4.4198	5.9085
	Recovery (%)	95.62–111.39	93.64–113.56	92.64–125.13	86.44–110.70	92.83–116.85	95.60–113.90	88.01–120.59
	R	0.9481	0.9279	0.7325	0.8185	0.9335	0.9200	0.8257
	p	≈ 0	≈ 0	0.0248	0.0070	0.0002	≈ 0	0.0061
Glycerol	N_m	14(5.23) ^e	102(30.56) ^e	85	84	124	96	842
	RMSEP (g/L)	0.2825(0.046) ^e	0.2873(0.013) ^e	0.6590	0.4739	0.3795	0.3207	0.5462
	Recovery (%)	94.89–104.39	94.81–105.69	87.95–109.82	91.36–106.54	93.94–105.12	94.46–106.10	95.01–112.23
	R	0.9516	0.9519	0.9628	0.8603	0.9351	0.9443	0.8402
	p	≈ 0	≈ 0	≈ 0	0.0029	0.0002	≈ 0	0.0046
Methanol	N_m	20(7.47) ^e	36(6.92) ^e	85	110	46	96	842
	RMSEP (%)	0.0141(0.002) ^e	0.0150(0.020) ^e	0.0532	0.0332	0.0255	0.0350	0.0322
	Recovery (%)	93.18–115.71	94.36–113.34	23.84–123.13	94.26–143.74	90.12–127.64	80.87–130.90	86.88–138.62
	R	0.9472	0.9464	0.8452	0.7237	0.8515	0.8164	0.6789
	p	≈ 0	≈ 0	0.0041	0.0275	0.0036	0.0073	0.0443

^a In CARS, $n_f = 10$; Fold(K) = 5 in Cross Validation; Number of MC sampling runs= 50.^b In UVE, $n_f = 12$.^c In IPLS, $n_f = 10$; Intervals= 10.^d In MWPLS, $n_f = 10$; Window size= 31.^e The standard deviation (σ).^f In LSSVR, $\gamma = 10^6$; $\sigma = 2$.

completely distinguish four classes of wine from four geographical origins. However, if it is found that the number of retained variables is decreasing, it will lead to the loss of sample effective information, and therefore four groups cannot be thoroughly classified. Therefore, selecting the proper number of variables can both decrease the

complexity of the model and enhance the explaining ability of the model. A trade-off is required between the smallest number of reserved variables (n) and the classification effect to decide the staying or going of the variables. As a result, a relatively small n was chosen on the basis of achieving an accurate classification result, and the reserved variables

are marked with red dots in Fig. 1 (IR spectra: $n = 96$; chromatography: $n = 92$; mass spectrometry: $n = 138$). It indicates that relatively few and dispersed variables were selected in IR spectra and chromatography. In mass spectrometry data (Molenaar et al., 2022), only a small number of variables are abandoned, probably related to the large number of characterized mass spectrometry peaks corresponding to the different groups in the complex mixture system.

In the published literature, it is found that the same red wine data was also classified by full PLS-DA (partial least squares discriminant analysis) and PLS-DA-based wavelength selection algorithm (Zhang et al. 2019b). However, these algorithms did not obtain satisfactory results. The classification accuracy of full PLS-DA, BSS (single bootstrapping space shrinkage), EBSS (termed ensemble of bootstrapping space shrinkage), GA-PLSDA (voting genetic algorithm-PLSDA), and s-PLSDA (sparse-PLSDA) was 59.3%, 60.0%, 71.1%, 60.4%, and 66.8%, respectively. The proposed G-based VS method, therefore, provides a good alternative for the classification analysis of small sample datasets.

4.3. The quantitative prediction results

4.3.1. The results of G-based VS-LSSVR model

LSSVR model was utilized for the prediction of multiple principal quality parameters of red wine with the selected characteristic variables. The prediction results including RMSEPs, recovery ranges, and determination coefficients R and p values of the prediction set were recorded. Through comparisons, IR spectra dataset offers superior performance in prediction accuracy than the other two datasets (chromatography and mass spectrometry). This is probably attributed to the fact that the wine samples contain various organic and inorganic compounds, and most of the fundamental frequencies of chemical bond vibration occur in the mid-infrared region, and thus the content of the substances is mainly reflected by the absorption wavelengths of the IR strong intensities. The quantitative results of IR spectra combined with the LSSVR model are given in Table 3, and the model intermediate parameters including σ and γ been used at their optimized values (in the footnote of Table 4).

It can be clearly seen from Table 4 that almost all of the prediction results with characterized variables ($N_m=96$) are preferable or equivalent to the prediction results with the full variables ($N_m=842$). Better recovery ranges, higher R values, and smaller p values were obtained, except for the slightly poor results of methanol (Recovery: 80.87~130.90%, R : 0.8164). It demonstrates that the G-based VS method not only simplifies the model but also enhances the prediction ability of the LSSVR model.

4.3.2. The comparisons of G-based VS and other variable selection methods

Conventional variable selection methods including CARS, UVE, IPLS, and MWPLS were also used for the IR spectra dataset, all of which were combined with the LSSVR model for prediction. Then the 9 quality parameters in red wine were predicted, and the RMSEPs, recovery ranges, determination coefficients (R), and p were recorded in Table 4. Obviously, the selected variables of CARS and UVE vary in each operation, so the mean RMSEPs with σ and mean numbers of selected variables (N_m) with σ of 100 repeated runs were calculated; whereas the selected variables of IPLS, MWPLS, and G-based VS are invariable, accordingly, the results are given for only one run. All of the model intermediate parameters including N_m and n_f (number of the principal factors of PLS) etc. have reached their respective optimized values during the prediction process (noted in the footnote of Table 4).

It can be seen from Table 4 that G-based VS-LSSVR obtains much better prediction results compared with IPLS-LSSVR. Most of the prediction results of G-based VS-LSSVR are superior or equivalent to those of MWPLS-LSSVR. In comparison with UVE-LSSVR, it is found that most of its mean RMSEPs, recovery ranges, and R values are equivalent to those of G-based VS-LSSVR except for in the prediction of methanol. However, the standard derivation (σ) value ("0.015 (0.020)") in the UVE-LSSVR prediction results of methanol is relatively large, which

indicates the fluctuation of the model. Furthermore, a few N_m values such as "571", "480" and "414" are much larger than those of G-based VS-LSSVR. On the whole, G-based VS-LSSVR selects fewer key variables for constructing the more concise model. Finally, through comparison, most prediction results of CARS-LSSVR are better than those of G-based VS-LSSVR model. Nevertheless, through further investigation, both the number and location of the reserved variables of CARS varied considerably over 100 runs, which can be reflected from $N_m(\sigma)$ in Table 4. Therefore, the prediction results of CARS may have overfitting performance (Li et al., 2019), which leads to the decline in the model robustness and credibility.

In our previous studies of variables selection (Ma et al., 2020), FD-VS (Fisher discrimination-variable selection) method was proposed and used; its prediction results are also presented in Table 4. It shows that, in general, G-based VS-LSSVR achieved equivalent results to those of FD-VS-LSSVR with a smaller number of feature variables.

In summary, the consensus G-based variable selection strategy provides a new criterion for evaluating and extracting the importance/key variables, which is feasible and effective for the simultaneous prediction of multiple wine quality parameters.

5. Conclusions

A classification and quantitative prediction modeling of wine were constructed using multi-type measured data coupling with variable selection. A new variable selection strategy named consensus G-based variable selection was combined with qualitative and quantitative chemometric models, and the prediction results show that it not only simplifies the model but also enhances the feature extraction and separation ability of the model. Therefore, the proposed method provides an effective variable selection tool not only for FTIR spectral quantitative analysis to predict multiple quality parameters of wine but also for classifying the wine from different geographical origins with IRs, chromatography and mass spectrometry data. Based on this, the performance of applying the G-based VS method to other complex samples or systems needs to be further investigated.

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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